ANISH AMBRETH

+971 50 138 2386 ◇ United Arab Emirates ◇ ambrethanish@gmail.com
LinkedIn ◇ Website

SUMMARY

Machine learning researcher focused on robust predictive modeling, time-series forecasting, and applied ML systems, with experience across quantitative research, LLMs, and federated learning.

EDUCATION

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), UAE — *Aug 2024 – May 2026*
M.Sc. in Machine Learning — GPA: 3.95/4.00

- Fully funded scholarship from the UAE government
- Thesis: Adversarial Robustness in Federated Learning

Indian Institute of Technology (IIT) Hyderabad, India — *Aug 2020 – May 2024*
B.Tech in Mathematics & Computing and Engineering Sciences — GPA: 3.42/4.00

PUBLICATIONS

1) **Anish, A.***, Tameem, B.*, Nils, L. (2026). Collaborative Threshold Watermarking. *Accepted for publication at ICML 2026. (∗ equal contribution).*

2) **Anish, A.**, Naveen, K., Mohsen, G. (2026). Client-Level Defense Placement for Adversarially Robust Federated Reinforcement Learning. *Accepted for publication at TMLR.*

UNDER REVIEW

1) **Anish, A.**, Naveen, K., Mohsen, G. (2026). Strategic Client-Level Defense Placement in Mobile Federated Learning. *Under review at MobiHoc'26.*

2) **Anish, A.**, Naveen, K., Mohsen, G. (2026). OTPA: Optimal Transport-Based Model Alignment Poisoning in Mobile and Edge Federated Learning. *Under review at MobiHoc'26.*

TECHNICAL SKILLS

- **ML**: PyTorch, scikit-learn, time-series forecasting, predictive modeling, federated learning, adversarial robustness, optimal transport
- **Core**: NumPy, pandas, SQL, order book data, FX modeling, feature engineering, statistical modeling, model validation
- **Tools**: Docker, FastAPI, Git, Jupyter, Weights & Biases, RAG pipelines, LangGraph, OR-Tools

PROFESSIONAL EXPERIENCE

ML Quantitative Research Intern – **Vatic Labs** (UAE) — *Jan 2026 – Apr 2026*

- Built a Python research pipeline for mining predictive alpha factors from raw order book data, including data cleaning, feature generation, model training, and historical evaluation.
- Designed a multi-agent factor-generation workflow to propose candidate signals, then evaluated them using linear models, TCNs, and LSTMs across historical forecasting tasks.
- Integrated generated factors into an end-to-end systematic research workflow, enabling faster iteration from signal hypothesis to model evaluation and backtest-ready analysis.

ML Quantitative Research Intern – **Standard Chartered** (UAE) — *Jul 2025 – Aug 2025*

- Built predictive models for FX rate dynamics across PKR, INR, and NPR, using market and competitor pricing features to support return-sensitive pricing decisions.
- Engineered and evaluated time-series feature sets from pricing, spread, and market movement data, testing model robustness across currencies and market regimes.
- Designed validation workflows to compare model performance, feature stability, and pricing sensitivity, improving interpretability for business and risk stakeholders.

Research Intern – **Dubai Health** (UAE) — *Apr 2025 – Jul 2025*

- Developed and deployed a **lightweight snoring detection** algorithm on ESP Sense for sleep monitoring, using signal processing and ML to infer events directly on-device.
- Achieved **99% reduction** in storage/processing and **50% lower** power consumption compared to phone-based app approaches, enabling scalable, cost-effective deployment.
- Worked with clinicians and engineers to **align model behavior with medical requirements**, presenting findings in an accessible, non-technical way.

Data Science Intern – **ADNOC** (UAE) — *Jan 2025 – Mar 2025*

- Built forecasting and scoring models on heterogeneous time-series and textual data to rank emerging technologies.
- Designed evaluation metrics and validation schemes to reduce noise and bias in model-driven rankings.
- Automated feature extraction and scoring pipelines, reducing manual analysis time by 90%.

Machine Learning Intern – **Proshort** (India) — *Jun 2024 – Aug 2024*

- Fine-tuned and quantized LLMs (LLaMA-3-8B, Phi3-mini-128k-Instruct) in PyTorch, tracking experiments with Weights & Biases to systematically optimize quality and latency.

- Deployed models using **Docker** and **FastAPI**, integrating RAG pipelines to improve response quality and accuracy by **30%**, evaluated with ROUGE-2 and human feedback.
- Designed and optimized **LangGraph-based workflows** and user session management, improving conversation continuity and robustness in production.

PROJECTS

Quantitative Reasoning for Text-to-Image Generation

- Curated a benchmark dataset to evaluate **quantitative reasoning** (counting, proportionality, fractions) in text-to-image diffusion models.
- Fine-tuned Stable Diffusion XL with **LoRA** and **DPO**, improving quantitative faithfulness in generated images; analyzed performance across multiple prompt categories.

Quantitative Analysis of Time Series Methods Performance

- Modeled and forecasted volatility and price dynamics of Gold, S&P 500, and Bitcoin using ARIMA, LSTM, XGBoost, and Prophet.
- Engineered lagged, rolling, and volatility-based features; evaluated models using MAE, RMSE, and stability across regimes.
- Analyzed bias–variance tradeoffs and robustness of models under non-stationary conditions.

POSITIONS OF RESPONSIBILITY

Graduate Assistant – MBZUAI (UAE) *Aug 2025 – Oct 2025*

- Assisted faculty with course and lab delivery, including content preparation, assignment support, and grading logistics.
- Helped organize and evaluate examinations, ensuring consistency and fairness across students.